INFORMATION TECHNOLOGY (SCIENCE – 97)

STANDARD XII — PRACTICE / PRELIMINARY QUESTION PAPER

Textbook-Based Question Paper • Science Stream

Marks	80	Time	_____
Student Name	_____	Seat / Roll No.	_____

Instructions

• All questions are based on the Maharashtra State Board Standard XII Information Technology textbook.

• Read every question carefully. For multiple-answer questions, select exactly the number of alternatives specified.

• Q.7: Answer any FIVE of the eight questions. Each carries 2 marks.

• Q.8(A): Attempt any ONE HTML program. Q.8(B): Attempt any ONE JavaScript program.

Q.1 — Fill in the Blanks [10 × 1 = 10 Marks]

1. HTML5 provides a new input type such as _____ for selecting a date.
2. The HTML element used to embed an audio file is the _____ element.
3. An image with several clickable regions is called an _____.
4. The logical structure of an HTML/XML document is represented by the _____.
5. JavaScript is a lightweight, platform-_____ scripting language.
6. The statement commonly used to leave a loop immediately is _____.
7. PHP is mainly a _____**-side** scripting language.
8. In PHP, a variable name begins with the _____ symbol.
9. IOT stands for _____ of Things.
10. The paperless exchange of business information using electronic means is called _____.

Q.2 — State whether the following statements are True or False [10 × 1 = 10 Marks]

1. HTML5 supports advanced form controls for collecting and validating user input.
2. The element can use multiple elements.
3. In a client-side image map, the element defines clickable regions.
4. JavaScript is case-sensitive.
5. Math.ceil() returns the nearest integer greater than or equal to a number.
6. The getDay() method of the JavaScript Date object returns the month number.
7. PHP variables begin with the $ symbol.
8. PHP code can be embedded inside HTML.
9. 5G is a generation of cellular network technology designed for high-speed data communication.
10. E-governance means using information technology in government processes and services.

Q.3 — Multiple Choice Questions: Select ONE correct answer [10 × 1 = 10 Marks]

1. Which HTML element is used to provide a video on a webpage?
a) <audio> b) <video> c) <media> d) <movie>

2. Which attribute of <img> points to a client-side image map?
a) map b) usemap c) hotspot d) region

3. Which JavaScript object represents the browser window?
a) document b) screen c) window d) browser

4. Which JavaScript statement is a multi-way decision statement?
a) switch...case b) repeat c) select...loop d) choose

5. Which extension is normally used for a PHP program file?
a) .html b) .php c) .js d) .css

6. Which PHP function returns a portion of a string?
a) trim() b) substr() c) count() d) splitOnly()

7. Which of the following is a cloud service model?
a) SaaS b) HTML c) DOM d) CSS

8. Which generation of cellular technology is the fifth generation of mobile networks?
a) 2G b) 3G c) 4G d) 5G

9. Which is an example of G2C e-governance?
a) Government-to-Citizen service b) Government-to-Employee only c) Business-to-Business d) Customer-to-Customer

10. Which type of e-commerce is a transaction between a business and a customer?
a) B2B b) B2C c) C2C d) C2B

Q.4 — Multiple Choice Questions: Select TWO correct answers [10 × 2 = 20 Marks]

1. Which two are valid HTML5 input types?
a) date b) email c) webpage d) browser

2. Which two tags are used for HTML5 media?
a) audio b) video c) soundbox d) film

3. Which two are parts of a client-side image map?
a) map b) area c) hotspotbox d) pointer

4. Which two are JavaScript built-in objects?
a) Date b) Math c) BrowserPage d) WebFile

5. Which two are JavaScript event handlers?
a) onFocus b) onBlur c) onJump d) onRun

6. Which two are valid PHP data types?
a) Integer b) Array c) VarcharHTML d) WebObject

7. Which two are types of PHP arrays?
a) Indexed b) Associative c) Browser d) Platform

8. Which two are cloud service models?
a) SaaS b) PaaS c) HTMLS d) DOMS

9. Which two are common e-commerce payment methods?
a) E-wallet b) Bank transfer c) HTML form d) CSS rule

10. Which two are forms/types of e-governance?
a) G2C b) G2E c) B2C d) C2B

Q.5 — Multiple Choice Questions: Select THREE correct answers [2 × 3 = 6 Marks]

1. Which THREE are features/characteristics of JavaScript?
a) It supports event-driven programming b) It is platform independent c) It provides built-in objects d) It requires compilation before every browser execution e) It cannot handle dates

2. Which THREE are types of PHP arrays?
a) Indexed b) Associative c) Multidimensional d) Browser e) Universal f) Graphic

Q.6 — Match the Following [4 × 1 = 4 Marks]

A	B
1. JavaScript Date object	a. Paperless exchange of business information
2. PaaS	b. Platform with tools for developing/hosting applications
3. EDI	c. Works with dates and time
4. G2C	d. Government services for citizens

Q.7 — Answer Briefly: Attempt ANY FIVE [5 × 2 = 10 Marks]

1. Explain any two advantages of using HTML5 advanced form controls.

2. Differentiate between the audio and video elements in HTML5.

3. Explain the purpose of the DOM in JavaScript.

4. Write any two JavaScript event handlers and state when they are used.

5. State any two features of PHP.

6. Explain the difference between client-side and server-side scripting.

7. Explain any two applications of IOT or 5G.

8. Explain any two advantages of e-commerce or e-governance.

Q.8 (A) — Write a Program Using HTML [5 Marks]

Attempt ANY ONE.

1. Create a webpage titled **"My College"**. Use a heading, horizontal line, short paragraph, an ordered list of three courses and an unordered list of five facilities.

2. Create a **Student Registration** webpage using an HTML5 form. Include name, email, date, number, URL and submit controls using suitable HTML5 input types.

OR

Q.8 (B) — Write a Program Using JavaScript [5 Marks]

Attempt ANY ONE.

1. Write an event-driven JavaScript program that accepts two numbers and displays their addition, multiplication, division and remainder.

2. Write a JavaScript program that accepts a string and counts and displays the number of vowels in it.

END OF QUESTION PAPER

Textbook basis: Standard XII Information Technology (Science) topics and exercises from Advanced Web Designing, Advanced JavaScript, Emerging Technologies, Server-Side Scripting (PHP), and E-Commerce/E-Governance.